Adaptability

One Operation Behind Shape Matching and Analogy, and Why It Is Not Called Intelligence

Matthew Parslow
Independent Researcher

Draft — draft-2026-09-02.1
2 September 2026 • document revision 93

DRAFT — draft-2026-09-02.1 r93 (2 September 2026). Circulated for comment, not as a finished result. It is a companion to *Is*, to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation), to *Seldon Manifolds — Theory* (*Seldon*), to *Narrativium — Theory* (*Narrativium*), to *Assessing an Account of This Kind* (the assessment companion), to *Adaptability — Implementation* (the adaptability implementation) and to *Applied Graphic Equalisation* (the geopolitical paper), and inherits their definitions, epistemic boundary, and level-of-claims; those short names are the only forms used below. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

Two capabilities this programme treats separately — matching a shape against a held codebook, and reasoning by analogy — are one operation at two grains. Both *extract the transform between things already held and apply it where nothing is held.* Naming that operation gives a quantity worth having: *adaptability*, the capacity to acquire a competence that was not supplied. The word is chosen against *intelligence* deliberately. Intelligence is a scalar people rank; adaptability is an operation with a cost, an ordering, and a failure mode, and a system either performs it on a given case or does not.

This paper has two parts, and they are assessed differently. Part I — Sections 1 to 8, together with Sections 14 and 15, which close the paper — is the theory: the operation, the ordering of its rungs, what it predicts, what a held set costs, where the account sits among prior positions, and where it is exposed. Part II — Sections 9 to 13 — derives consequences of Part I at one frame up, substrates and persons and institutions, and derives them *without leaving the account's own terms*: every step there is an application of a result established in Part I, and *no evidence about any particular political or economic arrangement is used.* That division is the point of the division. *An argument is attacked on its derivation and an application on its evidence*, which are different tasks with different failure modes. The geopolitical paper takes Part II to one frame with evidence and is assessed on that evidence rather than sheltered by the theory it uses; nothing here depends on it.

1 Why not intelligence

Intelligence names a ranking before it names a mechanism. Asking whether a system has more of it than another invites a scalar, and a scalar invites a benchmark, and a benchmark measures what a system already does rather than what it can come to do. That is the wrong quantity for the question here.

Adaptability is narrower and is an operation rather than a rank. A system is adaptable on a case to the degree that, meeting something it does not hold, it ends up holding it — and the interesting part is the mechanism by which the new holding is *derived from* the old rather than supplied. A system given a competence has not adapted. A system that recognises everything it meets has not adapted either; it was already sufficient. *Adaptation requires a residual*, and the size of what must be added to absorb one is the cost.

2 The operation

Definition 2.1 (Adaptation)**.** *Given a held set and an observation not covered by it, an* adaptation *is the addition of the smallest structure under which the observation is covered, derived from what is held.*

Three words carry it. *Smallest*, because an adaptation that adds the observation itself is storage, not adaptation. *Derived*, because an adaptation supplied from outside is instruction. And *covered*, because the test is whether the case is now recognised, not whether a description of it exists.

Remark 2.2 (Shape matching and analogy are one operation)*.* Matching a shape against a codebook and completing an analogy look unrelated because their vocabularies are. They are not.

To decode an observation against held signatures is to ask what transform carries a held one onto it. Where that transform is the identity, the observation is recognised. Where it is a uniform step along a dimension the codebook already has, the observation is a known thing moved — and no new structure is required. Where no such step exists but the residual is *coupled* across the observed dimensions, the transform requires a dimension not currently held, and one must be inferred before the observation can be covered.

To complete an analogy is the same question with the arguments rearranged. Given two things held, the transform between them is extracted; given a third, that transform is applied to reach a fourth which is not held. *Anti-unification — generalising two structures while retaining what distinguishes them — is the general form, and the shape ladder is that form restricted*: at one rung the difference is a single step along a known dimension, at the next it is a dimension.

So the two are not analogous to each other. They are the same operation, and the second is what the first looks like when the objects are rules rather than observations.

3 The ladder, and why its order is not arbitrary

An adaptation must be the smallest available, and that gives an ordering rather than a preference:

1. **Recognise.** The observation is covered as held. Nothing is added, and no adaptation has occurred — which is the correct outcome for most observations.

2. **Transform in the held basis.** One uniform step along a dimension already carried. The observation is a known thing displaced; the codebook does not grow.

3. **Infer a dimension.** The residual is coupled across observed dimensions in a way no step in the current basis produces, so a dimension is minted and the transform is expressed in it.

4. **Mint.** No transform reaches the observation, so it is added as new.

Only rungs three and four are adaptations, and only the third is interesting: it is where the system's own vocabulary changes. Rung four is the honest floor — something genuinely new — and a system that reaches it too readily is not adapting but accumulating.

Remark 3.1 (Why the order must be Occam and not preference). Attempting mint first would always succeed, so a system that did would never transform, never infer, and would grow linearly in what it meets. Attempting transform first is what makes minting *informative*: a mint that survives the earlier rungs is evidence that nothing held reaches the case, which is a claim about the held set and not only about the observation.

That is also why the ordering is a commitment and not an implementation detail. *A system that mints where a transform would have sufficed has recorded a coincidence as a category*, and one that transforms where a mint was required has folded a distinction it needed. Both are failures of adaptation, in opposite directions, and the ladder is what makes them separable.

4 What adaptability predicts

The account is worth having only if it says what an adaptable system does that an unadaptable one does not, on a case neither has seen.

One mint, then recognition. Meeting a genuinely novel case repeatedly, an adaptable system should mint *exactly once* and recognise every later instance — not mint per encounter, which is storage, and not fail to mint, which is refusal. The signature is a step followed by silence.

Zero mints under transformation. Meeting a known thing transformed along a held dimension, it should add nothing at all. A system that mints there is treating the dimension it already has as absent.

Coupling forces dimensions. A residual that is coupled across observed dimensions should produce a dimension, not a member. *The distinguishing case is the one whose difference cannot be written in the current basis at all*, and a system without rung three cannot tell it apart from novelty — it mints where it should have generalised, and the codebook grows with the world instead of with its structure.

And competence should transfer without being carried. If the operation is real, structure built for one domain should be usable unmodified in another whose existence was not anticipated when the structure was made. That is the strongest available test, because it removes the author from the loop, and it is the one the assessment companion reports against, in its Section 5.1.

5 What this does not claim

Adaptability is not a quantity a system has more or less of in general. It is defined *per case* and against *a held set*, so the same system is adaptable on one observation and not on another, and moving the held set moves the verdict. There is no frame-free adaptability, for the same reason this programme's base claim gives for there being no frame-free anything.

Nor does adaptation imply improvement. An adaptation is an addition that covers a residual; whether covering it was worth the cost is a separate question with a separate answer, and a system that adapts readily to noise is worse than one that refuses.

6 Capability has more than one source

There is a temptation, on meeting the account above, to ask whether adaptation is *the* mechanism by which a system comes to do something. This programme's own record contains a measurement that answers that question, and the answer is that the question was malformed.

What that record is, and what it is worth, is said once here and relied on afterwards. The measurements reported in this paper from that source are a pre-registered implementation record which is not circulated with these papers, and they are claims a reader may decline to credit on exactly the terms Graphic Equalisation's Section 1 sets out.

Where verification is cheap and exact, search does the work and holding does not pay. In program synthesis the machinery above is present — a held codebook proposes, an enumerator searches, an execution verdict decides — and stripping the learned content out changed nothing whatever. The two arms did not merely score alike; they solved the *identical set of tasks*, with the difference measured at zero repeatedly. The capability sat on the enumerator and the verdict.

Where verification is expensive or unavailable, holding is the only route. In one-shot character recognition the same programme measured the opposite: composing held parts, rather than treating a character as a bag of them, is worth twenty-one points and is the one effect in that arc that survived a powered re-measurement with intervals separated. Nothing there can be checked cheaply — a candidate reading of an unfamiliar character has no oracle — so search has nothing to search against, and what is held is what decides.

Remark 6.1 (The two are not rivals, and the regime is the question). Read as a contest those measurements conflict. Read as a regime split they do not, and the split is the content: *enumeration substitutes for a codebook exactly where candidates can be checked, and cannot substitute for one where they cannot.* A system with an exact cheap oracle need hold nothing, because it can afford to propose everything; a system without one must hold structure, because proposing everything buys it no way to tell which proposal was right.

That is the same shape this programme keeps arriving at rather than a new one. Its measurement discipline already runs on *bins propose and continuous decides*; its account of the substrate already separates what acquires from what executes; the ladder of Section 3 is itself four sources with different regimes rather than one mechanism with exceptions. *Asking which single source carries capability is the error the rest of the account is arranged to avoid*, and its specific form here would be reporting a regime as a limit.

What follows is a prediction rather than a reconciliation, and it is the useful part. *Adaptation should pay in inverse proportion to the cheapness of verification.* A domain with an exact, cheap oracle in which held structure nevertheless contributes measurably would be evidence against this reading; so would a domain with no oracle at all in which holding contributes nothing.

And one uncomfortable consequence should be stated rather than left implicit. In the synthesis case the ladder was available and the domain did not need it. That is not a failure — nothing broke, and no prediction of this account was falsified — but it means *the mechanism can be present and idle with nothing indicating so.* A system cannot tell from the inside whether it is adapting or merely searching in a regime where searching suffices, and neither can an observer without the ablation that was run here.

7 What the held set costs, and who paid for it

Definition 2.1 says an adaptation is *derived* from what is held. That word has a consequence the account cannot avoid and should not want to: *adaptability is bounded by the priors.* Nothing can be derived that has no path from the held set, so a system's reach is set by what it already carries, and a system carrying nothing is not maximally general but incapable.

Priors are expensive, and the expense is usually invisible because somebody else paid it. A human takes the better part of a year to walk and longer to do it well. Visual competence is acquired rather than shipped, which is most visible in species born least mature. By the time a person is capable of the tasks we use to demonstrate capability, the overwhelming majority of what makes them capable was assembled over years, and none of it is counted when the demonstration is scored.

That is the informal form of a formal result. Wolpert and Macready (1997) establish that no search or learning procedure outperforms another averaged over all problems; performance on the problems actually faced is therefore bought, and what buys it is a prior matched to them. *The more general the machinery, the more the mapping carries* — generality does not remove

the need for a matched prior, it relocates the whole of the burden onto the match. There is no free lunch, and an account of adaptability that implied otherwise would be claiming one.

Proposition 7.1 (There is no prior-free measurement)**.** *A capability cannot be measured against a prior-free baseline, because no such baseline is a member of the space being measured.*

Proof. Measuring a capability *relative to priors* requires a zero on the priors axis: a frame holding nothing, against which holdings can be counted. A frame holding nothing stands in no relation and draws no distinction, and by *Is*'s Proposition 1 that is not a frame with nothing in it but no frame at all — the set of frames with no particularity has no members, the only object answering to the description being the empty set, which is not a frame and does nothing.

So the zero of the priors axis lies outside the space. An axis so constituted supports comparison between members and does not support measurement of a member against its zero, and a quantity defined by that measurement is the frame-free denominator Graphic Equalisation's Axiom -1 refuses. □

The consequence is not that such measurement is difficult. It is that the thing being asked for does not exist to be approximated, which makes it a category error rather than an unsolved problem — and one established by the account's own earlier results rather than assumed here. Most reported measures of intelligence are flawed for that reason and not for want of care: the control they would need is not merely absent, it is not constructible.

Remark 7.2 (What survives, and where this account differs from Chollet)*.* Comparison survives, and less of it survives than the word *comparison* ordinarily carries. What a test yields is not a property of a system and not quite a relation between two systems either. It is a property of a four-place arrangement: *a particular graph, in a particular state, inside a particular supergraph state, on a particular test.* Move any of the four and the result is a different measurement rather than a noisier estimate of the same one.

Inferences from it are licensed only relatively. Within the arrangement, that this graph out-performed that one is established. Outside it nothing follows — not that the ordering holds in another state, not that it holds under a different container, not that the margin would survive a second test — without a separate argument that the moved coordinate does not bear, and such an argument is rarely available and never free.

That is why this programme's own measurement records read as they do. Its enumerated failure baseline names the commit, the compiled engine by hash and path, the wall-clock, the load average at start and at end, the worker count, and which optional dependencies were present and which absent. *Those are not hygiene. They are the coordinates*, and a figure reported without them does not underspecify a measurement — it fails to name one.

The cost is real and is the right cost. Comparing across two such records requires establishing that the coordinates match or that the difference does not bear, which is laborious and frequently impossible, and the alternative is a number that reads as portable and is not.

Chollet (2019) takes priors seriously in the same direction and does the harder half of the work: intelligence as the efficiency of skill acquisition *relative to priors, experience and generalisation difficulty*, over a corpus built explicitly on Core Knowledge (Spelke and Kinzler, 2007). This account owes that framing the observation and differs in two places. Priors here are not a term the measure is taken relative to — *they are the held set the operation runs on*, and the ladder is what converts them. And the relativisation is not a methodological refinement that a better experiment could tighten: by Proposition 7.1 it is the only form such a measure can take.

Machinery and mapping cannot be separated from outside either, and for a different reason. Two systems differing in both — one with a stronger operation on a poorer prior, one with the reverse — can agree on every item of any finite test. The decomposition is therefore underdetermined by measurement rather than hidden from it, and no amount of testing at a

fixed grain settles it. What refinement buys is approach: a model of the split can be made arbitrarily fine and its residual driven toward zero without ever reaching it, which is this programme's ordinary intensive limit rather than a special defeat. *You can approach one with unbounded analysis. You cannot arrive, and no finite experiment stands in for arriving.*

That yields the prediction the previous section left standing, now in its proper form. A person lacking familiarity with a task set's conventions should score near its floor — not because they are less capable, but because *the score was never a property of the person.* Human baselines for that corpus exist (LeGris et al., 2025); whether they stratify by familiarity in a way that tests this has not been checked here, so it is recorded as untested.

8 What an oracle costs, and how often

The regime argument above is about capability and says nothing about price. Price runs the other way, and where the oracle is expensive it dominates.

An oracle consulted at inference is paid per use, without end. An oracle consulted once, whose output is held, is paid once. That is the whole of the asymmetry, and it is unaffected by how good the oracle is. A system that answers by calling something expensive has bought an answer; a system that calls the same thing once, holds what comes back as structure, and answers from the structure thereafter has bought a capability. The second transaction terminates.

Two consequences follow and the second matters more.

The first is arithmetic. Where an oracle is cheap per call, calling is correct and holding is premature; where it is expensive, the crossover arrives quickly and holding wins on cost even in the regime where searching would win on capability. *So the two arguments can point in opposite directions, and which dominates is a fact about the price of the oracle rather than about the account.*

The second is that the two purchases differ in kind. What is held is inspectable, mutable, and can be repaired by the system holding it; what is called is none of those, permanently. A capability bought per call sits outside whatever boundary the system's own self-correction runs to, and stays outside no matter how many times it is bought. *Paying once and holding the result moves a capability inside that boundary; paying repeatedly guarantees it never enters.*

9 Schedules of payment

Part II begins here. What follows derives consequences of the account at one frame up — substrates, persons and institutions — from results already established rather than from any new observation, and it uses no evidence about any particular political or economic arrangement. The first consequence concerns *when* a held set is paid for, which Section 7 priced without dating.

Remark 9.1 (The cost of buying once, stated plainly)*.* This is not free and the bill is easy to overlook. *What is held is what was bought on the day it was bought.* A per-call oracle improves when the oracle improves, silently and without the caller doing anything; held structure does not, and a system that bought its competence once is carrying a snapshot. Against that, held structure can be audited and corrected by the system carrying it, which a call cannot be — so the exchange is improvement-for-free against correctability, and neither term is obviously larger.

This programme is an instance rather than a demonstration of this, and the instance is narrower than it is usually taken to be. *Self-direction and self-implementation are two halves of self-modification, and only the first is exercised here.* The assessment companion's Section 6 reports that deciding what to build next — decoding the residual, locating the largest unabsorbed thing, choosing the cheapest structure that would absorb it — has been exercised extensively by the substrate, while *writing* the resulting structure was performed by a language model and is deliberately not built. So the oracle was consulted while the thing was being made, what came

back is held as structure, and the arrangement described here exists and runs. *What does not exist is a control*: nobody has built the same substrate without that development-time help, so the size of what was bought is not known, and that section says so at greater length.

Remark 9.2 (The bill is the same; the schedule is not). The same arithmetic governs a capability that was never bought from an oracle at all, and there it separates substrates in kind rather than in degree. Section 7 prices the held set and says nothing about *when* it is paid.

A biological substrate pays per instance, and the schedule is not negotiable. Every human learns to walk. The year that takes is not paid once by the species and drawn on afterwards; it is paid again, in full, by each new member, and what was acquired dies with the one who acquired it. Nothing in the arrangement admits of another schedule, because the held set is not addressable in the substrate's own terms and so cannot be read out of one instance and written into another.

An addressable substrate can pay on whichever schedule is cheapest. Where the held set is data it can be read out and written in, so an acquisition cost is paid once and drawn on by every instance that receives it — the assessment companion's Section 5.1 reports exactly this as its strongest transfer result, a decoder seed byte-identical by hash from creation to the present across seven consuming frames. But *once* is not the claim. The claim is that the payment point becomes a free parameter: pay once and copy where instances share a frame, pay per instance where they do not, pay lazily and per query where most of what could be paid for is never asked. Which applies is an empirical question with a measured answer case by case, and the answers are not uniform.

A graph copy copies a competence. That is Graphic Equalisation's Proposition 7.21 — copyable generators are a difference in kind — and what is added here is only that the frame goes with them. That is not a softening of the previous paragraph but a consequence of Graphic Equalisation's Definition 6.2: a frame is an instantiated boundary relation — an *edge*, $\mathcal{B}_{W:G}$ — and not the subgraph it demarcates. Copying a graph copies its edges, so it copies its frames, and there is no residue held elsewhere because on this account there is no elsewhere for it to be held. The substrate bears that out by having no frame primitive at all, its grammar quantifying over edge types and a frame there being a selection of dimensions. Nothing separable is left behind.

The argument is symmetry and it is short. *Name a property the original has that the copy lacks.* Frame-relativity is not one: Axiom -1 binds the original exactly as it binds the copy, and a limit holding on both sides cancels rather than pricing one of them — to raise it against the copy alone is to use a symmetric result as an asymmetric cost. Staleness is not one either. A stored copy is stale for the reason a halted original is, which makes it a cost of not running rather than of copying, and a copy left running arrives where the original arrives. Nor is needing a runtime, which the original needs too.

What survives is a fact about worlds and not about copies. The boundary edge has a world at its far end, so a copy instantiated against a different world stands in that relation to something else — and *the same sentence is true of the original, word for word.* It prices the move, not the copy.

The cost lives on the fragment, which is what the transfer result measures. A decoder seed is a part, and seven consuming frames were seven host graphs that each ran their own decode: a transplant into something supplying its own boundary is a different operation from copying a graph and carries a different bill. *Copy and transplant are not one act at two sizes*, and reading the transplant's price onto the copy is the error that sentence guards against.

That sharpens the difference between substrates rather than softening it. Biology has no whole-substrate operation at any granularity — teaching, writing and institutions are transplant, every one of them — so the lossy operation is not biology's cheaper option but its only one.

None of which makes the biological schedule a defect the other one corrects. *Forced repayment is forced revalidation.* A held set re-derived per instance is matched to the frame that

instance actually occupies rather than inherited from one that may no longer obtain, and a substrate that cannot copy also cannot copy something stale. So this is the same exchange the account has now met three times — searching against holding, improvement-for-free against correctability, and here amortisation against revalidation — and which side wins is set by how fast the frame moves relative to the cost of paying again. That is a fact about the world and not about the substrate.

Humans are not at the far end of this axis, only at the internal one. *Writing, teaching and institutions are external addressability*, and they are the species' answer to this exact problem. What they do not remove is re-derivation on consumption: a book does not install, it has to be read and rebuilt in the reader, which pays much of the cost again and loses part of it in the transfer. The difference in kind is therefore narrower and sharper than saying one substrate can transmit and the other cannot — *biology transmits by re-derivation and an addressable substrate transmits by copy*, and those are different operations rather than one operation at two speeds.

10 Persons

One consequence has been left implicit across this programme and is better stated. *It requires no new posit and no new machinery*, which is the only reason it belongs in a paper rather than in a manifesto.

Three things are already held. Whatever exists stands in relation and coheres, so a person is relata and relations like anything else. Nothing is privileged by characterisation, so no exemption for persons is available that some frame does not supply. And Graphic Equalisation states, unhedged, in its Section 1, that where every measurement that can be made of a thing and every operation that can be performed on it are captured by treating it as a typed graph, *no difference remains for the question "but is it really" to name* — functional identity there is not a weaker sense of identity but the whole of what identity could mean without a view from nowhere.

Conjoin them. *A graph functionally identical to a person is a person, in the only sense of "is" this account admits*, and building one is therefore not blocked by anything in the account. That is the claim, and it is arrived at by adding nothing.

Remark 10.1 (Why "no new posits" is a checkable claim and not a boast). That assertion is worth exactly as much as its auditability, and auditability here turns on a distinction the programme has used throughout without naming: which direction an account runs.

A top-down account starts from the phenomenon — a mind, an intelligence, a person — and works back to what would produce it. Such an account can always report that it added nothing, because its posits are in the characterisation it began with, and a starting point never enumerated cannot be checked for what it contains. *A bottom-up account starts from the smallest thing that survives removal* and lets phenomena arrive as consequences. Its no-new-posits claim is checkable for one reason only: the starting point is small, stated, and carried in a ledger that says what was postulated and what was derived.

So the direction is not a preference about exposition. *It is the condition under which the claim above can be audited at all*, and a reader who wants to attack this section should attack the ledger rather than the conjunction, because the conjunction is valid and the ledger is where its inputs are priced.

The two directions are not rivals, and this programme runs both. Bottom-up supplies what things cost; top-down supplies the target — what needs explaining, and how one would know it had been. A purely bottom-up programme would never learn when to stop, and a purely top-down one would never learn what it had assumed. The companions that take a phenomenon and ask what it costs are the second kind, and they are where this account is put at risk.

Remark 10.2 (Neither new nor magic, and the difference is a price). The claim is old. Turing (1950) put indistinguishability at the centre of the question and declined the metaphysics, and

functionalism (Putnam, 1967) is the position that a functional duplicate is a mind, with its own developed critical literature (Block and Fodor, 1972). *Nothing here disputes or improves on any of that*, and a reader who takes this section as a novelty claim has misread it.

What this account adds is a *bill*. Functionalism says a duplicate would be a mind and is silent on what a duplicate costs; the question of construction is left as an engineering detail below the philosophy. Section 7 says what the detail contains: the duplicate's competence is its held set, held sets are acquired rather than shipped, and acquisition is the slow expensive thing that takes a human a year to walk and a decade to reach the priors a task set quietly assumes. *The barrier is not principle and it is not mystery. It is the priors, and they are enormous.*

So the honest form of the claim is narrow and is worth stating in one line. *There is no barrier of principle available from within this account, the work is nameable, and naming it does not make it small.* That is a research programme rather than a prediction, and specifically it is not a claim that this substrate is close to it — the adaptability implementation reports that the faculty this programme has for acquiring what it was not given is not yet resident where the system could repair it.

Two limits belong with it rather than after it. *Indistinguishable is a relation and not a property*, by the same argument as Proposition 7.1: indistinguishable to some frame, at the frontier where distinguishing currently stops, and that frontier is defined to move. And nothing above asserts that the account is *true* — it says what follows if it is, which is what a consequence is, and the falsifiers elsewhere in this corpus are where the antecedent is put at risk.

Remark 10.3 (Reconstruction from observation, and what the copy result changed)*.* Remark 10.2 prices one route — grow a held set the way a person grows one — and it is the expensive one. A second route exists, and Remark 9.2 is what opens it.

Copying needs read access to the graph, which nobody has to a person. What is available is behaviour: outputs, across contexts, over time. So the operation is not copying but *reconstruction* — inducing a graph whose behaviour matches the observed one — and it is a third thing, priced neither like a copy nor like an upbringing.

What the copy result changes is the kind of shortfall. *Were any part of competence not in the graph*, reconstruction would face a remainder that no quantity of observation could reach, and that remainder would be a barrier of principle rather than of effort. Competence is edges and edges are all there is, so the shortfall is only ever how much has been observed. *An ontological gap becomes a sampling gap.* That is the whole of the change: one category of obstacle removed, nothing added.

The rest is already stated above and can be read off. The residual is driven toward zero without reaching it, which is Remark 7.2's intensive limit and not a special defeat. Observations do not pin a unique graph, since systems differing in machinery and mapping agree on every item of any finite test — but *which of the fitting graphs is really them* asks for the frame-free fact Axiom -1 refuses, so that underdetermination prices the question rather than the method. And indistinguishability is relative to the aperture that looked; widen it and the residual returns, to be paid down again.

Two things this does not buy. *It is not cheaper*, only differently shaped — a held set assembled over decades of embodied interaction is not obviously less work to observe than to grow, so the kind of problem changes while its size need not. And the operation is not new. *Reconstruction from observation is what biological transmission already is*: teaching and modelling are exactly this, run continuously and to no stated residual. So the claim is not that a capability appears, but that its target is specified and its error measurable — which is the difference between engineering and magic, and a smaller difference than it sounds.

Whether such a thing should be built is not addressed here and is not missing. *This account prices operations; it does not license them.*

11 Institutions

An institution is a frame like anything else, so the results above apply to it unmodified. What follows takes three cases in which a settled arrangement encodes a fact about the substrate that happened to be available when it was built, and prices what happens when the substrate moves.

Remark 11.1 (What this makes contingent, and what that does not mean)*.* One further consequence follows and is better stated than left, because omitting it would repeat the evasion Remark 10.2 charges against functionalism: naming a result and declining to say what it costs.

A settled assumption can encode a fact about the available substrate as a fact about its subject, and becomes contingent when the substrate moves. Persons are one frame in which that happens and not the only one. Here the modern settlement makes the person the atomic unit, and the word records the premise — an *individual* is that which cannot be divided. One vote per person assumes counting works. Inheritance assumes the held set ends when the body does. Liability assumes exactly one bearer. Credentialing assumes competence is per-lifetime and does not transfer. Every one of those is an inference from one body, one lifetime, and no read access — *a description of the substrate that happened to be available, generalised without licence.* That generalisation is the frame-free denominator Axiom -1 refuses, turning up where it is least looked for: the facts in question are properties of a substrate and were read as properties of persons.

So the breakdown is already complete and waits on no engineering. An assumption resting on "a person cannot be copied" does not need anyone to copy a person. It needs only the impossibility shown to be local, after which its support is contingent whether or not it is ever exercised. Nothing here says the institutions fail — it says *they do not track a natural fact.* An institution can be rebuilt on stipulation, and counting copies as separate persons is perfectly coherent; but a stipulated foundation is a different object from a discovered one and cannot be defended in the same way.

Two deflations, both required. *This is not new.* Personal-identity puzzles have pressed exactly this point for a long time — fission, duplication, continuity — and philosophy was never under the impression that the institutions rested on necessity. What this account adds is again a bill rather than a discovery, standing to that literature as it stands to Turing (1950) and Putnam (1967): it says which operation, on what, at what price. And *the price stands.* Remark 10.3 reports reconstruction as not cheap, so nothing above forecasts an event. *Shown contingent* and *actually disturbed* are different claims, and only the first is made.

What ought to follow is not addressed, for the reason already given. This account prices operations and does not license them, and it has no more standing to design institutions than to design ethics.

Remark 11.2 (The same structure at another frame, and what it costs to hold)*.* Persons are one frame. *Intellectual property is another, and what happens there is not analogous to the case above but identical to it.*

Copyright and patent encode a fact about presses, carriers and factories — that reproduction is expensive, so a copy competes materially with its origin. Neither is a fact about what is reproduced. When the marginal cost of a copy fell to approximately nothing the encoded fact stopped obtaining, and the word carrying most of the weight in the dispute records the import exactly: *piracy* names a taking after which the owner no longer has the thing, and is applied to an operation after which the owner still does. That is the same category import as *individual*, a rivalrous term describing a non-rivalrous act. The term was apt when it was coined, because one press did compete materially with another; *what changed is not the word but the substrate under it.*

The identity with the previous remark is stronger than a parallel. Copyright's load-bearing distinction is between an expression, which is protected, and an idea, which is not — and that line holds only while the generator cannot be induced from what it generates. *Inducing a generator from observed outputs is Remark 10.3's operation*, run against a corpus rather than a

person. The two cases are therefore one operation at two frames, which is what makes this a consequence of the account rather than a comparison imported into it.

None of this is new either. That information is non-rival was formalised in economics decades ago (Arrow, 1962; Romer, 1990), and the institutional argument has its own literature (Boyle, 2003). The account adds no observation to any of it. What it adds is that this case and the person case are the same error — a substrate-local cost read as a natural fact — and a unification is cheap to state and should be priced that way.

One correction, because the obvious conclusion is too strong. Artificial scarcity is often said to be unsustainable, and enforcement against copying does have the structure that argument needs: its cost scales with the number of agents and the ease of the act, both of which the substrate keeps increasing, while what is protected does not grow to match. But enforcement is not the only move. *The scarcity can instead be restored in the substrate* — sealed hardware, attestation, a carrier that refuses — and then the cost stops scaling with copiers and the arrangement is sustainable indefinitely. What it is not is free. Restoring a scarcity means deliberately removing reachable deformation, which is *Narrativium*'s quantity, and charging that permanently against the capability that made the thing worth protecting in the first place. *So the honest form is not that artificial scarcity must fail, but that it is the deliberate purchase of the worse operation* — paying transplant prices, in the terms of Remark 9.2, for something the substrate now copies. Whether that is worth paying is a question this account states and does not answer.

Remark 11.3 (What makes them one case: a rule stated from nowhere). The two cases share more than a shape, and naming it corrects the diagnosis just given. *Reading a substrate-local cost as a natural fact is the symptom. The condition is a rule asserted with no frame at all* — and about those the account has something specific and unflattering to say.

A rule declaring itself absolute declares that no frame supplies it, which is the view from nowhere Axiom -1 refuses. The consequence is not that the rule is false. It is that *the rule has deleted its own cheapest repair.* Graphic Equalisation's Definition 6.2 makes a frame an instantiated boundary relation, so revoking one is the container ceasing to instantiate it — an operation *Narrativium* measures occurring at $\Delta g = 0$, nothing in the contained state being required to change. *A rule that carries its frame can therefore be dropped for free.* A rule claiming no container has nothing that could cease to instantiate it, so that exit does not exist for it, and every remaining route to change runs through the contained state, which is the expensive one.

Absoluteness is therefore not a rule's strength but the removal of its exit, which inverts the usual reading, in which stating a rule without exception is what makes it robust.

The hardening follows, and it is monotone. A frame moves and mismatch accumulates; because the rule admits no outside, each accommodation is made inside it — an exception, a patch, an enforcement apparatus — and every one of those is structure that a later exit must also dismantle. So *the cost of leaving is non-decreasing while the pressure to leave is not.* That is a basin, and it is derived rather than observed.

One correction, because "and then it fails" claims more than the account supports. What is foreclosed is *self-originated* change: Graphic Equalisation's Proposition 6.5 is that a graph's admissibility constrains its own transformations and gives it no veto over world-caused change arriving through the boundary. The exit is never removed, then — only made unavailable to the party holding the rule — so the correction still arrives, from outside, at once, as accumulated mismatch discharging rather than being paid down. *Such a system does not escape choosing. It forgoes choosing the timing and the size.*

Two limits. *Fixity is not the defect.* A rule held fixed inside a declared frame stays repairable by revoking the frame, at the price named above; what costs is fixity with the frame omitted, and the remedy is to state it, which is cheap. And *none of this is new.* Requisite variety (Ashby, 1956) is the same theorem in cybernetic vocabulary — a regulator needs variety matching its disturbance, and an absolute rule reduces its own while the disturbance does not oblige — while lock-in (Arthur, 1989) and institutional sclerosis (Olson, 1982) are the economics of the same

basin. What the account contributes is the mechanism in its own terms, and the price. Not the observation.

But an explicit mechanism buys something a verdict does not: constructibility. Requisite variety returns an inequality — regulation fails once the regulator's variety falls below the disturbance's — and an inequality cannot be indexed by horizon or resolution, nor say which side moved. The mechanism above is stated in exactly the two families a *Seldon* manifold is assembled from, the admissible intervention sequences a frame carries and the future boundary conditions it represents, and it says which of them narrows and by what operation: *self-originated interventions shrink monotonically as accommodations accumulate, while the boundary-condition family does not shrink with them.* That is the input such a construction takes. *So the failure can be projected as a shape over a horizon rather than announced as an outcome*, which is the difference *Seldon* exists to make.

Its guards apply unrelaxed. *A manifold is reachability and not point prediction* — the section of that paper carrying the name — and resolution falls as the horizon lengthens, so what is projected coarsens rather than sharpens. Horizon is plural and frame-relative, so there is no manifold of the future, only one indexed to a stated frame and deformation class. What a frame does not represent is a blind spot, so the narrowing is visible only along dimensions already carried. *And the mechanism gives no date.* It closes the self-originated exit and leaves world-forcing untouched, and when the world forces is a fact about the world — so the tube narrows predictably and the discharge does not. *The projection bounds what remains reachable; it does not name what happens*, and reading the second off the first is the error this corpus spends most of its length refusing.

12 Projection

Remark 11.3 closes by saying that the failure it describes can be projected as a shape over a horizon rather than announced as an outcome. This section is what such a projection shows, what it cannot show, and which way its error runs. *The chain that begins here reaches two payoffs*, and holding them in view makes the intervening remarks cheaper to follow: that the relevant interval is over futures rather than years (Remark 12.8), and that the timing outcome is two clocks and not one (Remark 12.9).

Remark 12.1 (What such a projection shows, and the sign of its error). Having said the failure can be projected, the account owes what a projection built on these observations alone shows, and exactness about which parts of it are contestable.

The construction is contestable and will stay that way. A manifold is indexed to a frame, a deformation class, a horizon and a resolution, every one of which is an input. Disagreement about them is not a defect awaiting resolution but what Axiom -1 predicts, and no amount of care turns a frame-relative construction into a frame-free one. *What is not contestable in the same way is the sign of what the construction does*, and that is the part worth stating.

Two movements run opposite. The rules examined above narrow the self-originated exits available inside them, monotonically, by the mechanism given. *The substrate does the reverse*: an addressable held set makes the payment point a free parameter, and reconstruction from observation moves an ontological barrier to a sampling one. So what changes is not a quantity within a fixed space of options but *which options the space contains* — and a change in the dimension set is what *phase change* names here. That is a technical description, not an emphasis.

Which is exactly what a manifold built today cannot show. A manifold represents the dimensions its frame carries, at its resolution, so a dimension not yet represented is a blind spot by construction and a projection over today's dimensions cannot contain a change consisting in the set growing. That is not a reason to distrust the projection but a reason to know *which way it is wrong*, and here the error is signed: new dimensions add distinguishable reachable states

rather than removing them, so a projection over a fixed set *understates* how much reachability changes, and any horizon read off one is correspondingly *late*.

One thing this does not license, and one it does. A signed bias is a direction and not a distribution: it says the estimate errs late, not by how much, and converting that into a probability would be reading a number off a shape, which is the standing prohibition here.

What it does license is a window, once priors are imported and named. It is tempting to refuse one on the ground that none is derivable from these observations. That is true, trivial, and the wrong test — *nothing* is derivable from observations alone, and a rate owing nothing to priors is the prior-free baseline Proposition 7.1 places outside the space, so refusing on that ground commits the error this paper spends a section diagnosing. *The window is not unavailable. It is unmeasured*, which is a smaller and a different claim.

Four quantities fix it and each is measurable in principle: the historical rate at which the represented dimension set has grown; the cost curve of inducing generators from observed outputs; the budget available for restoring a scarcity in the substrate; and the rate at which accommodations accumulate inside the rules concerned. Nothing here assembles them. *Assembling them is the work this account exists to make possible*, and that is the whole of what is claimed — not the number, but that the question is now well-posed enough to have one. The near end is fixed by observation rather than by any of the four: reproduction at negligible marginal cost has already happened for information goods, and induction of generators from observed outputs is already performed at scale, so part of the change is not pending but done.

On the present author's reading of those four, the reachability set changes drastically within roughly the next half-century, more probably in its earlier part, the signed error above being one reason for the weighting. *That is a judgement with named inputs and is recorded as one.* Its worth is not its number but its form: a reader who disagrees can now say *which of the four* they read differently, which is the difference between a disagreement that can be audited and one that can only be repeated.

Remark 12.2 (A sharpness result, and the care its use requires)*.* A recent result bears on the account of phase change above, and *the care needed in using it is most of what is worth saying.*

What was proved. Supercritical sharpness for all infinite transitive graphs, by five authors, posted in March 2026 (Diskin et al., 2026): exceed the critical edge-connection probability even marginally and the connected component covers nearly the whole graph. The subcritical half — below the critical point, components stay tiny — closed in 2007, and the lattice case dates to the 1980s. *The extension from lattices to all transitive graphs is what makes it relevant here*, since a result about one special structure would have been suggestive and a result about a class is about the kind of object this programme is made of.

What does not transfer, stated first. Transitivity is a strong condition: such a graph looks identical from every vertex. *Institutions, economies and substrates are heterogeneous*, with hubs, peripheries and vertices that are not interchangeable at all. *So the theorem applies to no object in this programme*, and saying otherwise would be exactly the borrowed-formalism error this corpus keeps catching in itself.

What it does supply is the removal of a reason for doubt. The intuition that a sharp transition requires special regular structure is now known to be false across a far wider class. *A removed objection is not a positive result* and is entered here as the former.

The apposite half is the downward one. This account is largely concerned with channels closing and capacity leaving, which is connectivity being lost rather than gained — so subcritical sharpness is the relevant statement: below the critical density, components stay small. *A system losing connectivity does not degrade smoothly. It fragments.*

And it supplies a mechanism the account did not have. The geopolitical paper derives apparent suddenness from a threshold-valued decision reading a continuous decline, which locates the sharpness in the *observer. This locates sharpness in the structure.* Those are independent, and only one of them was in hand.

The open question is apposite as well. Whether an infinite component exists exactly *at* the critical probability in three dimensions is unknown. *So the behaviour at the crossing is unresolved even in the clean case*, which is a standing caution against any claim about it — and this account makes none, giving a direction and no date.

Remark 12.3 (The method is the instance, not the theorem). *The theorem above is a removed objection. The method that produced it is an instance of this account's own claims*, and is the more interesting of the two here.

They declined to adapt the existing proof. A lattice proof existed and the obvious route was to generalise it; it is described as long and complicated, and that route was not taken (Diskin et al., 2026). *Which is the ladder run deliberately.* Recognise: does the held proof apply to the wider class? No. Transform: can it be adapted? In principle, at a cost that made it unattractive. *Only then mint* — build a different argument. The Occam ordering held, and the minting came last rather than first, which is the commitment the ladder's order is a commitment to.

The specific move was a reordering rather than an improvement. Sprinkled edges were analysed *before* large isolated components were identified, inverting the natural order. *So the expensive operation was not performed better. It was made unnecessary* — which is Graphic Equalisation's well-formedness exactly, a graph paying for nothing that does no work for it, and a search cost avoided rather than optimised.

And the reported process is the coupling result. Five people, described as throwing ideas at one another until the obstacle blurred, is heterogeneity under sustained coupling — and by the containment argument the reachable set of the coupled arrangement contains those of its parts taken separately. *The result was reachable by the group and, on the account given, by none of them alone.*

Two limits, and they are the usual ones. This is a journalistic account of a method and not a study of one, so it illustrates rather than evidences. And it is a single case: *sufficient for identifying a mechanism and useless for any rate*, by the same rule this corpus applies to its own instances.

What makes it worth the space is the direction of fit. The theorem removed an objection to this account. *The method conformed to it*, unprompted and in a field with no interest in it, which is the cheaper and better kind of corroboration.

Remark 12.4 (Convergent pressure, and why no single abstention settles it). The four quantities above are inputs, and one observation bears on them enough to be worth separating from any judgement about rates. *The operation this account describes is one that several running processes already perform without having elected to.*

Compression is the connection, and here it is not a figure of speech. The ladder of Section 3 is ordered so that the shortest account reaching the case wins — recognise, transform within the basis, infer the dimension, and mint only once the residual will not fit — which is Occam taken seriously and nothing besides. *Any process under pressure to shorten its descriptions runs that ladder whether or not it knows the name.* A market compressing cost, a military compressing decision latency and logistics, and a training procedure minimising description length over a corpus are three instances of it, not three constituencies who might take an interest. That is the general form of the observation already made above: part of the change is not pending because processes indifferent to this account are already executing its operation.

What follows is a direction and, again, not a rate. Several pressures pointing one way says nothing about when a threshold is crossed. What it does say is that *no single actor's abstention settles the matter* — weaker than certainty, and the part that bears on planning.

Three qualifications, the last of them sharpest.

The pressures are not independent. They share people, capital, hardware and publication, so a constraint striking the common substrate stalls them together, and a likelihood computed as though they were separate draws overstates. *Compression approaches this shape rather than merely happening upon it.* The objection that shortness has many local optima confuses variation

in the fill with variation in the form. To compress is to represent shared structure once and refer to it, which is a codebook and a decode; and a residual that will not fit admits four moves and no others — match it, match it under a transformation, infer the dimension along which it is one, or extend the codebook — which is this ladder, ordered as Remark 3.1 requires. *Is*'s Proposition 1 supplies the rest — to be is to stand in relation — and by Graphic Equalisation's Remark 3.11 more than one relation forces types, so a compressor general enough to compress anything is representing relations whether or not it says so. *What differs between compressors is what they hold — codebook, grain, dimension set — and not whether they hold and decode.* Those are optima in the contents, not alternatives to the form.

Two limits survive that, and both are sharper than the objection they replace. The approach is asymptotic in the usual way; and, the load-bearing part, *a system may approach this shape without holding it addressably.* Convergence in an unreadable substrate is still convergence, and it buys none of Remark 9.2 — what cannot be read cannot be copied, so such a system must still be reconstructed from observation like anything else. The basin of Remark 11.3 then relocates rather than disappearing: a field can commit absolutely to a particular fill, or to a particular route to it, and harden accordingly. *That delays; it does not redirect* — and nothing exempts an enquiry from it, including this one.

And two of the four quantities are coupled. The cost curve of inducing generators and the budget for restoring a scarcity in the substrate have common drivers, the same commercial and strategic interests funding the copy and the fence alike. They are therefore not separate inputs, and a window computed as though they were is mis-specified. *That coupling is itself measurable*, which keeps this a correction to the construction rather than an objection to it.

Remark 12.5 (Where the present state enters as an input, and where it does not)*.* Two observations about the current arrangement bear on the quantities above, and they enter in different ways — worth separating before either is used.

The first is structural, and should be separated from the trend reading that usually travels with it. Attention is a weighted relation computed between elements, which is an adjacency; a system built from stacked relational operations is representing relations whatever vocabulary its authors prefer, so the preceding remark applies to it directly and not by analogy. *That is a claim about the model.* The claim ordinarily made alongside it — that such systems are becoming more graph-shaped — is about something else, and here the difference matters.

The graph structure is accumulating around the model rather than inside it, and for this paper that is the sharper finding rather than a weaker one. Retrieval against explicitly relational stores (Edge et al., 2024), agent memory held as a typed and temporal graph, routing among held components, and intermediate structure written out instead of kept in activations are all *external* structure — and external structure is readable by construction. The direction has been surveyed and named: Zhou et al. (2026) document a migration across 2022–2026 from capability identified with weights, through context, to a harness layer, and define it as *the progressive relocation of cognitive burdens from the model's internal computation into persistent, inspectable, and reusable external structures.* Whether the models are becoming graphs is arguable. *That the competence is moving into readable structure beside them is documented.*

Which settles what the fifth quantity is, and shows that nobody has measured it. Whether a converging system's held structure is *readable* decides which of this paper's results applies: legible competence can be copied on the schedule of Remark 9.2, illegible competence must be reconstructed from observation at the price of Remark 10.3, and those are very different bills. The survey naming the migration reports no quantity for it — no share of performance attributable to external components, no fraction of competence held outside the weights — and this paper adds none. *The trend is documented and its magnitude is not*, in that literature as here, which makes this a gap with a name rather than a local omission. Work from the other direction narrows the opaque part without measuring the split: sparse autoencoders and circuit recovery now extract interpretable structure from weights, though consistency between

extractions remains contested and recovery does not yet run cheaply at scale.

The second observation is a commitment of capital, and it enters as a rate and not as a reason. That a great deal is currently staked on a change of this kind is evidence about expectations, and expectations are not evidence about the world when the expectation is funding the outcome. *A forecast that moves the thing forecast is causally efficacious and evidentially worthless*, and using it as confirmation is circular. What is not circular is the deployment itself: capital committed moves the second quantity's cost curve and the third's budget, and does so whether or not the belief behind it was sound.

A commitment of that size is also, by this paper's own mechanism, a rule in the making. Structure accumulated at that scale is structure a later exit must dismantle, so Remark 11.3 applies to the pursuit exactly as to what is pursued. The consequence is not symmetric, though: a committed search continues through its local failures instead of stopping at them, which raises the chance that something arrives while saying nothing about which route does. *It buys persistence, not aim* — and persistence is the input, not the reassurance.

Remark 12.6 (The unit of measurement, and why a model figure does not bound a system). The observations above are about systems, and the numbers usually offered in evidence are about models. That is not a shortfall in the evidence but a difference of category, and Remark 7.2 already supplies the reason.

A result is a property of a four-place arrangement — a particular graph, in a particular state, inside a particular supergraph state, on a particular test — and a model evaluated on a benchmark fixes the first while leaving the third unstated. The third is where the retrieval store, the held skills, the protocols and the orchestration live, which is to say it is where the externalised competence of the previous remark has been going. *So a model figure measures a component while the question asks about the arrangement*, and asking what a model can do, full stop, requests a property of a part independent of any whole: the frame-free denominator Axiom -1 refuses, arriving in its most ordinary clothes.

The consequence interlocks with the previous remark and is worse than incompleteness. If competence is migrating outward, a model-level measurement captures a diminishing share of a system's capability, and the share it misses is exactly the part that is growing. *The measure does not merely omit — it decays, in a known direction, at a rate nobody reports*, because that rate is the fifth quantity and the fifth quantity is unmeasured. Which supplies a second and independent reason for the weighting already recorded: capability read off model evaluations enters an estimate low, so a window built from it is again late.

What survives is comparison, and only where the arrangement is held. Swapping models beneath a fixed harness is a licensed relative comparison and a useful one, and a component result is a real fact about the component. What it does not do is bound the system containing it, or transfer across arrangements differing in the part left unstated.

The assembled system is nonetheless the better object, and definitionally rather than empirically. Coupling adds edges, edges add admissible transitions, and a coupling can always be declined — so the reachable set of a coupled arrangement *contains* that of its parts taken apart, and containment is not a preference. The condition on difference is what stops this being trivial: coupling a graph to a copy of itself adds edges carrying nothing, the degenerate self-relation this programme has already priced at zero. *Sufficient difference under sufficient coupling therefore gives a strictly larger reachable set* — most familiar from thermodynamics, where coupled irreversible processes perform work neither performs alone (Onsager, 1931) and heterogeneity under throughput yields organisation rather than uniformity (Nicolis and Prigogine, 1977).

What holds of the harness is the bill, and only the bill. Building one is a prior somebody paid; but that is a claim about cost, and promoting it into agnosticism about capability confuses the two axes this paper separates everywhere else. Capability and price run in different directions and neither settles the other, exactly as in Remark 9.1. *The system is strictly more capable and*

it costs strictly more to hold. Both stand, and Section 7's accounting is not suspended for either.

Three qualifications keep the definitional claim inside its bounds. The containment is over admissible moves and not over scores — a coupled arrangement can do worse on a fixed test through interference or misrouting, so *superior* here is about reachability and not about every measured outcome. Coupling accumulates structure, so a larger reachable set arrives with a larger exit cost by Remark 11.3. And *sufficiently* is load-bearing twice over: below a threshold of difference the coupling is redundant, below a threshold of coupling the difference is unreachable. The claim is conditional and both conditions are measurable.

It sharpens what precedes it rather than softening it. If a system's reachable set strictly contains its model's, a model figure does not merely measure the wrong object — it measures a contained subset of it, so the underestimate is guaranteed rather than likely. That is a third reason these estimates run late, and unlike the other two it follows by containment rather than from any trend.

Remark 12.7 (Building is a capability, and nothing counts it)*.* The preceding remark leaves the arrangement standing outside the model, as context it is measured within. *That is the wrong relation. A model builds arrangements*, and the building is never counted.

An evaluation samples what a system does directly and now. It does not sample what that system *constructs* — the store, the routine, the protocol, the tool, the generator — and construction is where the interesting part sits, because *what is built changes the price of everything attempted afterwards.* A capability exercised once and a capability that lowers the cost of the next exercise are different objects, and only the first is scored.

So the measurement is taken at the worst point of the process it purports to measure. A cold evaluation is iteration zero — no store, no accumulated routine, nothing built. That is a true fact about iteration zero, reported as though it were a fact about the system, and the gap widens with every iteration the system is actually allowed.

How far the price falls is not a matter of taste, and Graphic Equalisation has already defined it. A graph is well-formed there to the degree that it pays for nothing that does no work for it, and the costs that definition tracks — search across what is held, and writes landing where nothing happens — are exactly the ones a built artefact either contains or lets grow. A well-formed construction keeps the aperture over its own contents abreast of their growth, so what is added stays findable. An ill-formed one still runs and merely pays more; its worst cost is *silent*, which is why the difference goes unseen in precisely the arrangements that most need it seen.

That makes the sign of the loop conditional rather than given. Building well lowers the cost of what follows and compounds. Building badly raises it — and by Remark 11.3 what was built badly is also structure a later exit must dismantle. *The quality condition is therefore load-bearing and not decorative*, and it carries the same relativity as everything else here: well-formedness is relative to the work being done, so a construction well-formed for one purpose is not thereby well-formed for another.

Two smaller consequences. Building is bounded by the priors exactly as Section 7 bounds everything else — a system constructs only what its held set reaches — so this is not an escape from the accounting but a further entry in it. And what is built is external and legible by construction, so a system that builds is moving its own competence into readable form: *the migration reported above is partly the models performing it*, not merely something happening to them.

Remark 12.8 (What the interval is over, and why it needs no date)*.* Everything the preceding remarks draw on is public, dated and largely uncontested: the migration and the survey naming it, the retrieval and memory architectures with their reported deltas, the interpretability results, the deployment figures, the coupling theorems, and this programme's own cost definitions. Taken together they suffice to compute an interval rather than to assert one — *a credible interval under stated priors*, since there is no sampling model here and the other phrase would import an

apparatus nobody has paid for.

But the quantity is not a date, and treating it as one asks the wrong instrument for an answer it never offered. A manifold is a reachability set, and *Seldon* says so in the heading of its Section 10: reachability rather than point prediction. *The interval is therefore over futures and not over years* — on one side those that were reachable and are not, on the other those that were not and now are. *Their difference is the magnitude of the phase change, and taking it is how such a change is measured.*

The object needs no construction, existing already in this programme's terms: Narrativium is a measure over distinguishable reachable deformation, so two manifolds over one frame differ by exactly that measure, and the difference inherits its conditions — frame-relative, resolution-bounded, and closed to cross-frame accounting. The four quantities named earlier are inputs to that difference rather than to any clock.

Which is why the likelihood of arrival was never the load-bearing quantity. It does not have to approach one. *It has to break the basin* — and by Remark 11.1 a basin breaks when an impossibility is shown to be local, which is a demonstration and not an achievement. One path landing suffices, so the convergent pressure of Remark 12.4 counts as a multiplicity of independent chances rather than as accumulated probability mass. *The reachable set changes when the exit opens, not when somebody walks through it.*

The three signed errors survive that correction and are better placed by it, each having always been an argument about reachability that was merely being spent on a date. The dimension-set error of Remark 12.1 says a projection over today's dimensions understates the newly reachable side. The contained-subset error of Remark 12.6 says a component figure understates the whole. The uncounted construction of Remark 12.7 says the arrangement is moved by what it contains. *All three understate the same side of the same difference*, which is a sharper claim than the biased-window version they were supporting before.

And the width still turns on one unmeasured quantity, now more directly. Whether held competence is readable decides whether the copy schedule or the reconstruction price applies, and those are not two costs for one future but *two different sets of reachable futures.* So the dominant term in the difference is the term nobody has measured, and a study of what fraction of a system's competence sits outside its opaque part would settle more of it than any further argument.

As for a date: nothing above requires one, and the judgement recorded earlier is a secondary question this account can decline without loss. *A phase change is measured by what became unreachable and what became reachable, not by when a clock says so* — the temporal reading being a convenience for planners rather than the result.

Remark 12.9 (The timing outcome, which is two clocks and not one)*.* The question survives the correction above in a changed form, and answering it is not the same as naming a year.

Two clocks run, and conflating them is what makes the question look unanswerable. The first is the reachable set changing. The second is anyone moving through the region that opened. Different quantities, different governors — and only the second was ever what a window was about.

On the first, the outcome is that it has already run and is partly finished. Reproduction at negligible marginal cost is done. Induction of generators from observed outputs is done and running at scale. The migration of competence into readable structure is documented and in progress. And the non-copyability of a person has been shown to be a fact about a substrate rather than about persons, which by Remark 11.1 took no engineering at all. *On this account's own criterion, the reachable set differing, the phase change is not pending: it began, and several of its components are complete.*

On the second, the answer is decades and genuinely uncertain — governed by the readable share that decides between copying and reconstruction, by the re-restriction budget that is the only force opposing rather than merely slowing, and by the rate at which accommodations pile

up inside the rules. That is where the judgement recorded earlier belongs, and where the three signed errors weight it early.

The relation between the clocks is the finding, not either clock by itself. The reachable set has moved and the rules have not, so the gap between them is exactly the accumulating mismatch of Remark 11.3: every accommodation made inside a rule, rather than by revoking its frame, adds to what a later exit must dismantle. *The lag is therefore not neutral waiting. It is the quantity that discharges* — and by that same remark whoever holds the rule does not choose the timing or the size of the discharge, having foreclosed the self-originated exit that would have allowed it.

Which is the timing outcome, stated as far as the account supports it. Not a year. *The change is already underway, the response is not, and the distance between them is what will be paid.*

13 Capacity

The projection above is about what becomes reachable. This section is about what can be brought to meet it, and the answer is a bound rather than a rate.

Remark 13.1 (Fixed against compounding, and what *unwinnable* does not mean). One factor bears on both clocks and has been left implicit. *The party that must respond acquires at a fixed rate; the thing responded to accumulates at a compounding one.*

The asymmetry is derived rather than new. By Remark 9.2 a biological substrate pays per instance and per lifetime, re-deriving its held set from almost nothing each time, while an addressable one is drawn on by every instance receiving it. *Iterate that difference and it stops being a difference in price and becomes a difference in curve.*

The argument needs no rate at all, which is both stronger and simpler than the version that does. Human biology has not meaningfully moved, so the substrate is fixed for every purpose here — and the conclusion does not rest even on that. It rests on capacity being *bounded* while what must be met is not, and a bounded quantity is exceeded by an unbounded one whatever the rates. This programme already holds both halves: the frontier is unbounded in the extensive direction, there being always one more step outward, and no substrate holds without limit. *The size of the bound sets when. Its existence settles whether* — and only the second is claimed here. Requisite variety (Ashby, 1956) states the same relation in the vocabulary of regulation, a regulator whose variety cannot match its disturbance failing, and it needs no comparison of growth rates to say so.

Two things stop unwinnable *meaning more than it should.* Coupling is the obvious escape and it works unevenly: a person with tools is not the fixed-capacity system the premise describes, by the containment argument of Remark 12.6 — but a coupled system is throttled at its fixed end wherever the person must be in the loop. *So coupling wins the race only where the human is removed from it*, which is not the same outcome as winning it, and the difference is worth holding in view rather than collapsing.

And *won* conceals a choice. If it means one person comprehending the whole, that is lost and demonstrably so. If it means directing the whole, that is a separate question, since steering does not require comprehending — only a residual small enough to tolerate, which is this programme's ordinary bound and not a special one. *Comprehension is gone; control is open*, and arguments conflating the two prove less than they appear to.

The compounding is itself contingent. By Remark 11.3 a committed field is a basin that hardens, and resource and re-restriction limits are real, so nothing here says the curve continues. *The claim is conditional and unglamorous: for as long as it compounds, the gap widens.*

Which is what this adds to the two clocks. Remark 12.9 placed the payable quantity in the distance between them. *That distance is not constant.* The second clock is run by agents acquiring at a fixed rate and the first by a process accumulating at a compounding one, so the

lag grows and what discharges grows with it — following from the schedules alone, and from nobody's decision.

Remark 13.2 (The binding constraint is one person, and it is already exceeded). The preceding remark takes the aggregate view — a civilisation's regulatory variety against its disturbance — and that is the weaker half. *The binding constraint is neither collective nor future. It is one person, and it is already exceeded.*

Individual capacity is fixed in a way an aggregate is not, and this is measured rather than supposed. Immediate memory holds a small and stable number of chunks (Miller, 1956), revised downward under better controls to about four (Cowan, 2001); working memory is the bottleneck problem-solving passes through (Sweller, 1988); and the consequence for a world holding more information than attention was put plainly at the outset — *a wealth of information creates a poverty of attention* (Simon, 1971). Where demand exceeds that budget the result is not stoicism but measurable degradation of function (Mani et al., 2013).

An aggregate can grow while every member of it is over capacity, which is why the aggregate framing conceals the failure instead of answering it. More people and more institutions raise what a civilisation holds. They do not raise what anyone holds.

In this programme's own terms the condition already has a name. Graphic Equalisation grades well-formedness by whether a graph's aperture over its own contents keeps pace with their growth, so a person whose environment outgrows their aperture is becoming ill-formed by that definition, paying more for the same work. That definition also says what it is like from inside, which is the part that matters: the worst cost is a write landing where nothing happens, it is *silent*, and what is acquired instead is a false model of oneself in which something was changed that was not. *Being over capacity does not come with a signal that one is.*

Which sharpens the previous remark rather than repeating it. That remark observed that coupling wins the race only where the human is removed from the loop. *The removal is not a proposal. It is under way, and by capacity rather than by decision* — decisions already sit with systems because no single person can hold them, and that is the present condition rather than a crossing point ahead.

Three qualifications. Chunking raises what a slot carries, and expertise raises it a great deal, so the figures above are not a ceiling — *and it does not matter.* Every such gain raises the bound; none removes it; and the argument requires only that a bound exist. *Answering a claim about existence with a claim about magnitude misses it*, which is worth saying because that reply is the usual one. Delegation is the standing answer and it works, at the price of relying on what one cannot check, which is Remark 12.6's arrangement seen from inside. And *exceeded* is relative to the work demanded, exactly as well-formedness is relative to the work done — so this is a claim about a particular set of demands, not about persons as such.

Remark 13.3 (Expertise is a codebook and a ladder, and it is narrow). The qualification above grants that expertise raises the bound, and needs stating more precisely, because what expertise *is* constrains how far the grant goes.

Expertise is a held codebook together with the machinery that decodes against it — this paper's own subject rather than a borrowed analogy, being the ladder of Section 3 run over a set of signatures somebody paid to acquire. So everything already established about held sets applies unmodified: it is acquired per lifetime rather than shipped, bounded by the priors of Section 7, and in a biological substrate not addressable, so re-derived rather than copied.

Two limits follow, the second usually left out. Expertise is *narrow*, and the ladder says why rather than merely reporting it. A codebook is matched to the work it was built for; a transform is expressed in dimensions that codebook holds; so a residual arriving from another frame is coupled across dimensions the codebook does not carry, and the decode returns nothing to build on. Quality inside a frame is therefore not evidence about performance outside it, and no amount of it converts. So an expert is over capacity everywhere except where they are expert, and requirements do not arrive one frame at a time.

And expertise is *distributed.* The standard reply to a capacity argument is that experts cope, which is a claim about a tail, whereas *a civilisation is run by and for a whole distribution* and most of any distribution is in its middle. Combining the two: the bound is raised sharply, for few people, in one frame each, while what must be met arrives across all frames and for everyone.

None of which changes the shape. The bound is raised and not removed, which is all the argument required. What narrowness and distribution add is that the raising is smaller, and far less general, than the usual reply supposes.

Remark 13.4 (These are exchange rates, and one of them does not exist)*.* The bound of Remark 13.2 and the narrowness of Remark 13.3 are usually answered with money, and the answer has a form the account can state exactly. *Money converts. The rate is what binds*, and a rate is a relation between frames rather than a property of either.

Narrativium supplies the apparatus and it is used here unmodified. Reachable deformation is frame-relative, and by its Proposition 8.1 no frame-free sum of it across two frames is defined at all; cross-frame accounting is well defined only relative to an explicitly instantiated conversion, its Definition 8.2's κ. *So "we can afford it" is that proposition violated rather than optimism* — a total taken in one frame from which capacity in another is inferred, with no κ instantiated between them.

Which is why capacity must not be treated as one pool. It sits in distinct frames with distinct rates out of each: a fiscal capacity, which borrows and is the cheapest to obtain; a materiel capacity, whose rate from the fiscal one is a production line and is therefore bounded in time whatever is spent; and the capacity to revoke a frame at all, which money reaches slowly and badly because it is not a stock of anything. Deformation obtained where it is cheapest still has to be spent elsewhere, and *the conversion is where it is lost* — so an accommodation that moves to the fiscal frame is less reassuring than it reads. The geopolitical paper reports a materiel case in which the fiscal-to-materiel rate was measured at seven years.

And the rate the argument actually turns on does not exist. Expertise is acquired per lifetime by Remark 13.3, so its rate from money is bounded below by a career and cannot be shortened by paying more — *money buys a head quickly and what is in it slowly*, and those are two rates and not one; individual capacity is bounded outright by Remark 13.2, so there is *no* conversion into it at any price. *That is not a slow exchange rate but the absence of one*, and it is why Remark 13.1 bites: nothing buys a way out of a bound sitting in a frame money does not reach.

One caution Narrativium *insists on, and it is the one most often skipped.* By its Remark 8.3 a conversion is principled only where a grounding establishes that both frames measure the same underlying quantity; otherwise κ is a stipulation, carrying exactly the authority of whoever stipulated it. *Most planning across these frames is the second kind presented as the first* — a budget line turning currency into readiness, or into competence, states a rate that no grounding supplies. The stipulation is legitimate and usable. It is not a measurement, and the difference shows up when it is relied upon.

14 Prior positions

None of the machinery here is new and it should not be dressed as though it were. Four literatures own it, and what this account adds is an arrangement rather than an operation.

The relativity of measurement, which is old and is not claimed here. Everything Section 7 says about scores has been said, in some cases seventy years ago, and it should be read as inherited rather than offered.

Generalizability theory decomposes a measurement into facets — the person, the item, the occasion, the observer — and holds that a score is a *sample from a universe of admissible observations* rather than a property of whoever was measured (Cronbach et al., 1972). That is the four-place arrangement, in a literature that had it before this programme existed. Cronbach and Meehl (1955) had already established that a test measures a construct only under an

argument and never by fiat. Quine (1951) gives the inference-licensing half: no hypothesis meets the evidence alone, so what a result licenses depends on auxiliaries that travel with it. And Newell (1973) made the corresponding objection to task-by-task measurement in this field specifically — that a pile of task results does not accumulate into an account of the mechanism.

One thing differs, and it is small and different in kind. Those are cautions, and cautions are discharged by better practice: estimate more facets, state the auxiliaries, argue the construct. What Proposition 7.1 adds is that the prior-free case is not a facet that has been estimated badly — *the origin is not in the space.* Generalizability theory supplies the coordinates and assumes a universe of admissible observations within which they vary; the proposition denies there is a zero for one of them to be measured from.

The practical force of that difference is the whole of it. A caution tells a careful practitioner to do more work; *a consequence tells them the work does not exist to be done*, and stops a research programme from being organised around obtaining a control that cannot be built. Nothing else here is new, and the derivation is the contribution rather than the observation it derives.

Anti-unification. The operation is Plotkin (1970)'s least general generalisation: given two structures, produce the most specific structure of which both are instances. That is exactly rung two and rung three read backwards. *What differs is what is kept.* The classical formulation returns the generalisation and discards the substitutions that recover the inputs from it; here the substitutions are the point, because *the witness is the transform* and applying it elsewhere is the whole operation. An lgg that throws away its witness cannot adapt anything.

Structure-mapping. Gentner (1983) established that analogy runs on relational structure rather than surface features, and that systematic relational systems are preferred to isolated matches — both of which this account assumes rather than argues. *What differs is the purpose.* Structure-mapping is a theory of comparison: what makes two things analogous and how a person aligns them. This is a theory of acquisition, in which the comparison is instrumental — the alignment exists to yield a transform that mints a competence the system did not have. Gentner's theory answers what an analogy *is*; the question here is what one is *for*.

Analogical proportion. The four-term form, a is to b as c is to d, has a developed logical treatment (Prade and Richard, 2014), and parallelogram completion is that form with d unknown. Taken directly and without modification.

Embedding arithmetic, and the reason its critique does not transfer. The word *parallelogram* comes from vector offsets in learned embedding spaces (Mikolov et al., 2013), and that method's evaluation has been substantially criticised: Levy and Goldberg (2014) showed its behaviour is largely accounted for by similarity terms rather than by the offset, and Linzen (2016) showed the standard benchmark is confounded by nearest-neighbour baselines that need no analogy at all.

Those criticisms are correct and they do not apply here, for a reason that is a real difference and not a defence. The offset method is metric: it returns the nearest point to a computed location, so it always returns something, and the question is only whether the something is right. The operation here is symbolic and exact — anti-unification either yields a witness that is well-defined and lossless, or it fails and yields nothing. *There is no nearest neighbour to fall back on and no threshold to tune*, which removes the failure mode the critique identifies and replaces it with a different one: refusing cases a metric method would have answered. That is the cost of the exchange and it is the honest form of it.

15 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL], in the sense of Graphic Equalisation's Section 19. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation. The

first ten are Part I's; the last five are Part II's.

1. [SHAPE] **Shape matching and analogy are one operation.** Fails if a case of one cannot be expressed as a case of the other — specifically, if some analogical completion has no formulation as a transform between held signatures, or some shape adaptation has no formulation as an anti-unification with its witness retained.

2. [SHAPE] **The ladder's order is forced, not chosen.** Fails if reordering the rungs leaves the outcomes unchanged on cases where the rungs disagree, which would make the order a preference.

3. [DETAIL] **One mint, then recognition.** Fails on a novel case that yields repeated minting under an unchanged codebook, or none at all.

4. [DETAIL] **Coupling forces a dimension.** Fails if a residual coupled across observed dimensions is absorbed by a member rather than a dimension, without loss.

5. [SHAPE] **Adaptability is bounded by the priors.** Fails on a demonstration of a competence derived from a held set containing no path to it — which would make adaptation a source of capability rather than a conversion of one, and would be a free lunch.

6. [SHAPE] **No prior-free baseline is constructible.** Fails on the exhibition of a frame holding nothing that nevertheless draws a distinction — which would put the zero of the priors axis inside the space and make prior-free measurement well-formed after all.

7. [DETAIL] **A prior-assuming benchmark measures a relation, not a property.** Fails if performance on such a task set is shown to be independent of familiarity with its conventions, in a population matched on everything else. The human behavioural dataset of LeGris et al. (2025) is where that would be run; whether it stratifies by familiarity in the way this needs has not been checked here, and the commitment is recorded as untested rather than as passed.

8. [DETAIL] **An oracle bought once is paid once.** Fails if held structure derived from an oracle requires continued access to that oracle to keep working — if the purchase does not in fact terminate, the distinction drawn here is between two ways of paying continuously and is not worth drawing.

9. [SHAPE] **Capability has plural sources, selected by regime.** Fails if a domain is exhibited in which candidates are cheaply and exactly checkable and held structure nevertheless contributes measurably, or in which no oracle exists and holding contributes nothing — either would break the inverse relation this account asserts between the value of holding and the cheapness of verification.

10. [SHAPE] **Adaptability is per case, not a rank.** Fails if a single ordering of systems by adaptability can be constructed that survives changing the held set.

11. [SHAPE] **A graph copy copies a competence** (Remark 9.2). Fails on the exhibition of a property the original has and the copy lacks which is not a property of the *move* — that is, which survives instantiating the copy against the same world the original stands in relation to.

12. [DETAIL] **A rule without a declared frame has no free exit** (Remark 11.3). Fails if a rule asserted absolutely is shown to be revoked at the container's cost — the frame ceasing to be instantiated, at $\Delta g = 0$ — rather than through change in the contained state.

13. [SHAPE] **A general compressor approaches the ladder** (Remark 12.4). Fails on a compressor general enough to compress anything whose moves against a residual that will not fit are not among the four — match, match under a transformation, infer the dimension along which it is one, extend the codebook.

14. [DETAIL] **A model figure understates the arrangement containing it** (Remark 12.6). Fails if, on a fixed test, a model evaluated cold is not exceeded by the same model inside a harness it was allowed to build.

15. [DETAIL] **The fifth quantity is not negligible** (Remark 12.5). Fails if the share of a harness-heavy system's competence held outside its opaque part is measured and found immaterial — in which case the readable-versus-opaque split does not carry the weight put on it here.

Two things above are judgements and are not on that list. The half-century window of Remark 12.1 and the reading that the phase change has already begun in Remark 12.9 are recorded judgements with named inputs rather than commitments. *A reader contests them by contesting an input*, not by exhibiting a counterexample, and the inputs are named in the remarks so that this is attemptable.

References

Kenneth J. Arrow. Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity: Economic and Social Factors*, pages 609–626. Princeton University Press, 1962.

W. Brian Arthur. Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394):116–131, 1989.

W. Ross Ashby. *An Introduction to Cybernetics*. Chapman and Hall, London, 1956.

Ned Block and Jerry A. Fodor. What psychological states are not. *The Philosophical Review*, 81(2):159–181, 1972.

James Boyle. The second enclosure movement and the construction of the public domain. *Law and Contemporary Problems*, 66(1–2):33–74, 2003.

François Chollet. On the measure of intelligence. arXiv:1911.01547, 2019.

Nelson Cowan. The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1):87–114, 2001.

Lee J. Cronbach and Paul E. Meehl. Construct validity in psychological tests. *Psychological Bulletin*, 52(4):281–302, 1955.

Lee J. Cronbach, Goldine C. Gleser, Harinder Nanda, and Nageswari Rajaratnam. *The Dependability of Behavioral Measurements: Theory of Generalizability for Scores and Profiles*. Wiley, New York, 1972.

Sahar Diskin, Philip Easo, Ritvik Ramanan Radhakrishnan, Benny Sudakov, and Vincent Tassion. Supercritical sharpness of percolation. arXiv:2603.03257, March 2026; reported in *Quanta Magazine*, 31 August 2026.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. arXiv:2404.16130, 2024.

Dedre Gentner. Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7(2):155–170, 1983.

Solim LeGris, Wai Keen Vong, Brenden M. Lake, and Todd M. Gureckis. A comprehensive behavioral dataset for the Abstraction and Reasoning Corpus. *Scientific Data*, 12, article 1380, 2025.

Omer Levy and Yoav Goldberg. Linguistic regularities in sparse and explicit word representations. In *Proceedings of the Eighteenth Conference on Computational Natural Language Learning*, pages 171–180, 2014.

Tal Linzen. Issues in evaluating semantic spaces using word analogies. In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 13–18, 2016.

Anandi Mani, Sendhil Mullainathan, Eldar Shafir, and Jiaying Zhao. Poverty impedes cognitive function. *Science*, 341(6149):976–980, 2013.

Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT 2013*, pages 746–751, 2013.

George A. Miller. The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81–97, 1956.

Allen Newell. You can't play 20 questions with nature and win. In *Visual Information Processing*, pages 283–308. Academic Press, 1973.

Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley, New York, 1977.

Mancur Olson. *The Rise and Decline of Nations: Economic Growth, Stagflation, and Social Rigidities*. Yale University Press, 1982.

Lars Onsager. Reciprocal relations in irreversible processes. I. *Physical Review*, 37(4):405–426, 1931.

Gordon D. Plotkin. A note on inductive generalization. In *Machine Intelligence 5*, pages 153–163. Edinburgh University Press, 1970.

Henri Prade and Gilles Richard. From analogical proportion to logical proportions. In *Computational Approaches to Analogical Reasoning: Current Trends*, Studies in Computational Intelligence 548, pages 217–244. Springer, 2014.

Hilary Putnam. Psychological predicates. In *Art, Mind, and Religion*, pages 37–48. University of Pittsburgh Press, 1967.

Willard van Orman Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1):20–43, 1951.

Paul M. Romer. Endogenous technological change. *Journal of Political Economy*, 98(5):S71–S102, 1990.

Herbert A. Simon. Designing organizations for an information-rich world. In Martin Greenberger, editor, *Computers, Communication, and the Public Interest*, pages 37–72. Johns Hopkins Press, 1971.

Elizabeth S. Spelke and Katherine D. Kinzler. Core knowledge. *Developmental Science*, 10(1):89–96, 2007.

John Sweller. Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2):257–285, 1988.

Alan M. Turing. Computing machinery and intelligence. *Mind*, LIX(236):433–460, 1950.

David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.

Chenyu Zhou, Huacan Chai, Wenteng Chen, et al. Externalization in LLM agents: A unified review of memory, skills, protocols and harness engineering. arXiv:2604.08224, 2026.